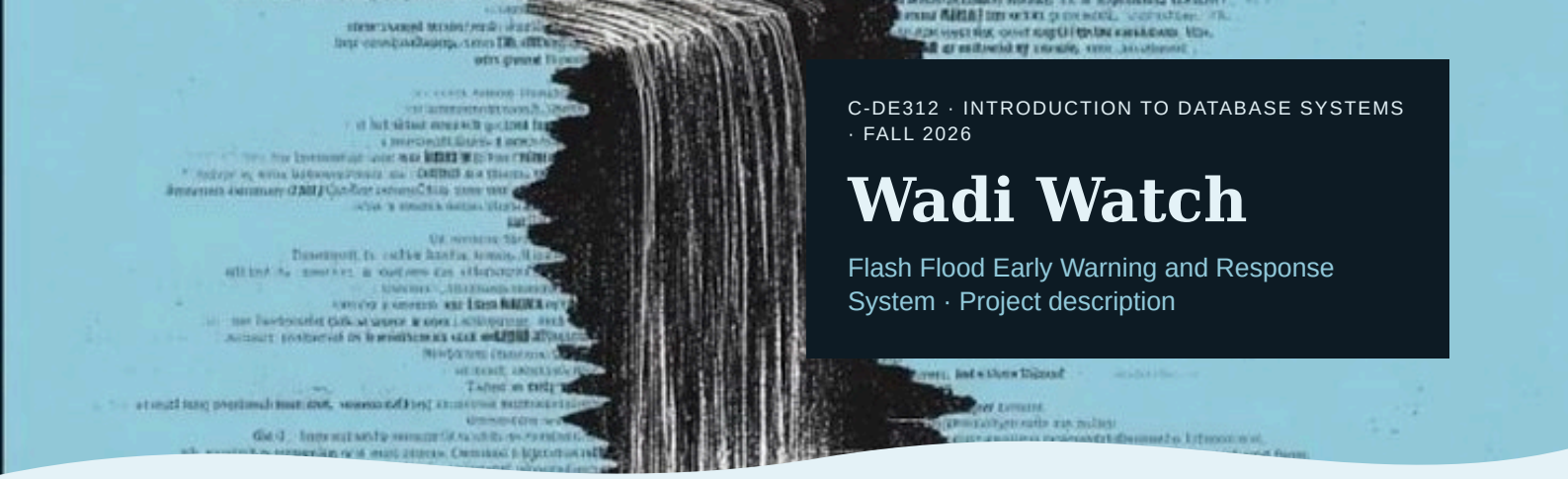


C-DE312 · INTRODUCTION TO DATABASE SYSTEMS · FALL 2026

Wadi Watch

Flash Flood Early Warning and Response System · Project description

About the project

Our team is a software company. The client is the **Red Sea and Sinai Civil Protection Directorate**, and the person who sent the request is Brig. Gen. Sameh Fouad, Director of Operations.

They asked us to build one system that collects flood information, warns people early, and helps organise the rescue.

The problem

> Flash floods hit towns in Sinai and along the Red Sea after heavy rain in the mountains. The water can reach a town in **less than an hour**. Today the sensors, the weather forecasts and the citizen phone calls are not connected, and the operations room works with phone calls and paper maps.

Our goals

- Put sensors, forecasts and citizen reports in **one system**.
- **Warn people early** in Arabic and English, even where mobile data is weak.
- Make sure **no alert is forgotten**: if nobody answers, it goes to the Regional Director.
- **Coordinate the rescue**: send the right team and keep track of every task.
- Write a **report after every flood** so the Directorate can learn from it.

What the system does

1. Sensing
Sensors send readings every 5 minutes. Forecasts come every 3 hours.

2. Risk levels
Every town and road gets a level: Normal, Watch, Warning or Emergency.

3. Alerts
At Watch, the Duty Officer has 10 minutes to confirm or dismiss, or it goes to the Regional Director.

4. Warnings
Arabic and English messages go out by cell broadcast, SMS, the app and loudspeakers (mosques, churches). At Emergency they are sent automatically.

5. Citizen reports
People send a photo and location, like "water crossing the road". Also works by SMS.

6. Rescue
The Dispatch Officer picks a team by distance, equipment (boats, 4x4, helicopters) and workload. Teams update from tablets.

7. Shelters
Shelters have capacity and supplies. Volunteers register arrivals so families can find each other. At 90% full, new people go to the next shelter.

8. Reports and drills
After each flood, a report shows warnings, response times and people rescued. Drills use fake data and never send real warnings.

Who uses it

ROLE	WHAT THEY DO
Duty Officer	Confirms or dismisses Watch alerts within 10 minutes.
Regional Director	Gets the alert if the Duty Officer does not answer in time.
Dispatch Officer	Assigns rescue requests to teams. Gets a reminder after 30 minutes with no update.
Rescue teams	Update their tasks from tablets.
Maintenance	Gets a message when a sensor is silent for 20 minutes.
Volunteers	Register people arriving at shelters.
Citizens	Receive warnings and send reports.

Rules and limits

- The rainy season starts in 7 months, so we must be ready before then.
- Mobile data is weak in the mountains, so reports must also work by SMS.
- The system must support drills on simulated data.
- Everything is saved in one database.

TIMING RULE	TIME
Sensor readings	every 5 min
Weather forecasts	every 3 hours
Sensor counts as silent	after 20 min
Watch alert must be answered	within 10 min
Task with no update	reminder at 30 min
Shelter nearly full	90% of capacity

Our database design

We designed one relational database for Wadi Watch, and we will build it with SQL in SQL Server Management Studio (SSMS). The sensor readings arrive every 5 minutes and grow very large, so they go in their own table that is split by time. The database has four zones, like a building with four floors:

1. The map
Things that exist: sensors, towns, roads, teams, shelters.

2. Measurements
Things that are sensed: readings, forecasts, sensor outages.

3. The incident
Things that happen: alerts, warnings, reports, rescue tasks, shelter arrivals.

4. Memory
What we want to know: history log, event reports.

Why this design? Every fact is stored once, so the data stays consistent. Old records are never overwritten, so we can always see what happened and when. The database can also answer the client's four questions:

- How long from the first sensor alarm to the first public warning?
- How many rescue requests did each team handle, and how fast?
- Which shelters were full, and which had space?
- Which sensors fail most often?

What we still need to find out

- How do other dry countries, such as Oman and Jordan, run flash flood early warning?
- What is cell broadcast and how is it different from normal SMS?
- What standards exist for emergency alerts, such as the Common Alerting Protocol (CAP)?

Our team has three members, so the optional damage assessment and volunteer management modules are not part of our project.

Use case

This part shows who uses Wadi Watch, what they do, and how the entities work together in each use case. The actors are the ones in the table "Who uses it", plus the sensors and the weather service, which send data by themselves.

Entities in the system: Town, Road, Sensors, Reading, Weather Forecast, Risk Assessment, Alert, Warning, Officer, Flood Event, Event Report, Citizen, Citizen Report, Team, Equipment, Rescue Task, Task Status Update, Shelter, Supply, Person, Family, Volunteer, Shelter Registration.

Use cases and how the entities interact

NO.	USE CASE	ACTOR	ENTITIES	HOW THEY INTERACT
UC1	**Send sensor readings**	Sensors (automatic)	Sensors, Reading, Town	A sensor sends a reading every 5 minutes. The system saves it as a Reading. If a sensor is silent for 20 minutes, Maintenance gets a message.
UC2	**Get weather forecast**	Weather service (automatic)	Weather Forecast	A new forecast comes every 3 hours. The system saves the rainfall prediction, the location and the source.
UC3	**Calculate risk level**	System	Reading, Weather Forecast, Risk Assessment, Town, Road	The system uses readings and forecasts to give each town and road a level (Normal, Watch, Warning or Emergency).
UC4	**Confirm or dismiss alert**	Duty Officer, Regional Director	Alert, Officer, Town, Road, Flood Event	At Watch, an Alert is created for a town and a road, and it belongs to a Flood Event. The Duty Officer has 10 minutes to confirm or dismiss it, or it goes to the Regional Director.
UC5	**Send warning**	System	Warning, Alert, Citizen	The system sends a Warning in Arabic and English by cell broadcast, SMS, the app or loudspeakers. At Emergency it is sent automatically.
UC6	**Send citizen report**	Citizen	Citizen, Citizen Report	A citizen sends a photo and location (also by SMS). It is saved as a Citizen Report with its type and urgency.
UC7	**Assign rescue task**	Dispatch Officer	Citizen Report, Rescue Task, Officer, Team, Equipment	The Dispatch Officer makes a Rescue Task from the report and picks a Team by distance, equipment and workload. After 30 minutes with no update he gets a reminder.
UC8	**Update task status**	Rescue team	Team, Rescue Task, Task Status Update	The team updates the task from a tablet. Each update is saved as a Task Status Update.
UC9	**Register people in shelter**	Volunteer	Volunteer, Shelter Registration, Shelter, Person, Family	The volunteer registers each person who arrives, with the family and the arrival time. At 90% full, new people go to the next shelter.
UC10	**Manage shelter supplies**	Volunteer	Shelter, Supply	Each shelter has supplies with their quantities, so everyone knows what is available.
UC11	**Write event report**	System, Directorate	Flood Event, Event Report, Alert, Warning, Rescue Task	After each flood, an Event Report shows the warnings, the response times and the people rescued.